Actual results that are much better than estimated could be an indication of skilled and quality execution, whereas actual results that are much worse than estimated could be an indication of inferior execution quality.

The difference between actual and estimated, however, could also be due to actual market conditions during trading that are beyond the control of the trader—such as sudden price momentum, or increased or decreased liquidity conditions. (These are addressed below through the use of the z-score and market adjusted cost analysis.)

The pre-trade performance benchmark is computed as follows:

$$\textit{Pre-Trade Difference} = \textit{Estimated Arrival Cost} - \textit{Actual Arrival Cost} \tag{3.20}$$

A positive value indicates better performance and a negative value indicates worse performance.

Since actual market conditions could have a huge influence on actual costs, some investors have started analyzing the pre-trade difference by computing the estimated market impact cost for the exact market conditions—an ex-post market impact metric. While this type of measure gives reasonable insight in times of higher and lower volumes, on its own it does not give an adequate adjustment for price trend. Thus investors also factor out price trend via a market adjusted performance measure.

Index Adjusted Performance Metric

A market adjusted or index adjusted performance measure is intended to account for price movement in the stock due to the market, sector, or industry movement. This is computed using the stock's sensitivity to the underlying index and the actual movement of that index as a proxy for the natural price appreciation of the stock (e.g., how the stock price would have changed if the order was not released to the market).

First compute the index movement over the time trading horizon:

$$\textit{Index Cost}_{bp} = \frac{\textit{Index VWAP} - \textit{Index Arrival Cost}}{\textit{Index Arrival Cost}} \cdot 10^4_{bp} \tag{3.21}$$

Index arrival is the value of the index at the time the order was released to the market. Index VWAP is the volume weighted average price for the index over the trading horizon. What is the index volume weighted price over a period? Luckily there are many ETFs that serve as proxies for various underlying indexes such as the market (e.g., SPY), or sectors, etc., and thus provide easy availability to data to compute volume weighted average index prices.